7.3 Activities: fractions

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Fractions are much easier when you can touch and see them. Keep the strips from Activity 1 in an envelope and bring them out whenever fractions come up.

1. Paper fraction strips

You need: 6 strips of paper the same length (A4 cut lengthways works), scissors, coloured pencils.

How: Leave one strip whole and write "1" on it. Fold the next in half and label each part 1/2. Fold others into quarters, eighths, thirds, sixths (for thirds, roll it into an S shape and flatten). Colour each strip differently and cut the pieces apart (or keep them whole and lay them on top of each other).

Use them to: find equal fractions (which pieces cover exactly 3/4?), compare (is 2/3 more than 3/4?), and add (lay 1/2 and 1/4 end to end: what single strip matches?).

Talk about it: "Which pieces can make exactly 1/2? Why can't thirds make 1/2?"

2. Pizza and pancakes

You need: a real pizza or pancakes on a pancake night, a knife.

How: Cut a pancake into quarters and another into eighths. "If I eat 1/4 and you eat 3/8, who ate more? How much is left altogether?" With two and a half pancakes on a plate: "How many halves is that?" (5.)

Talk about it: "Why is it 2 1/2 and also 5/2? Which way of saying it is easier when you're hungry?"

3. Fraction wall

You need: squared paper (or the strips from Activity 1).

How: Draw a "wall": a row 24 squares long labelled 1, then rows split into 2, 3, 4, 6, 8 and 12 equal parts (24 works for all of these). Use it to answer questions like "1/3 + 1/4 = ?" by finding pieces of one size that fit both (twelfths).

Talk about it: "Why did we pick 24 squares? What if we'd picked 10?"

4. Recipe maths

You need: a simple recipe that uses fractions (1 1/2 teaspoons, 3/4 litre of milk, *een halve liter melk*), a measuring jug and spoons.

How: Make one and a half or double the recipe together. He works out each amount and measures it. "3 × 1/4 litre of milk. How much is that?"

Talk about it: "Did you add, or multiply? Does it matter?"

5. Folding paper: fraction × fraction

You need: square sheets of paper, two coloured pencils.

How: Fold a square into quarters one way, and colour 3 of the 4 strips (3/4). Now fold it in half the other way and shade half of the coloured part with the other colour. Unfold. How many small

rectangles are there? (8.) How many have both colours? (3.) So half of 3/4 is 3/8. Try 2/3 of 3/4 the same way.

Talk about it: “Can you guess a shortcut from the pictures? Does it always work?”

6. Fraction war

Play “Fraction war” or “Fraction strips race” from GAMES.md. The strips from Activity 1 help him check who wins each round.

Conversation prompts for any day

- “Is 3/4 a number? Where would it go on a number line from 0 to 2?”
- “Which is more: 2/3 of a big pizza or 3/4 of a small one?” (It depends, and that’s the point.)
- “Can you think of a fraction between 1/2 and 1? And between 1/2 and 3/4?”